



SOPHIS: an open-access database for quantitative studies of contemporary philosophy of science

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Abstract

Quantitative approaches have proven highly effective in studying the structure and dynamics of science, but their success depends on the availability of reliable and comprehensive data. In the humanities, existing data sources remain fragmented and limited, hindering systematic research. In this paper, we introduce SOPHIS, an open-access database specifically designed for the quantitative study of contemporary philosophy of science. The current version of SOPHIS covers eight core journals in philosophy of science over the period 2005–2019. SOPHIS integrates information on epistemic contents, their material carriers, and the individual and collective actors involved in their production and dissemination, thereby offering a unified framework for mapping the epistemic and social landscape of the field. We illustrate its potential through three exploratory applications: (i) identifying key actors and analyzing the distribution of prestige, (ii) examining the factors of prestige accumulation, and (iii) uncovering the fine-grained community structure of the discipline. SOPHIS represents a step toward building a richer data infrastructure for the humanities, enabling transparent, large-scale, and evidence-based analyses of their social and epistemic dynamics.

Keywords Quantitative studies of philosophy · Philosophy of science · Bibliometric data sources · SOPHIS · Sociology of philosophy · Graph databases

Introduction

Data sources play a central role in bibliometrics and quantitative studies of science. The citation indexes created sixty years ago by Eugene Garfield spurred the large-scale development of citation analysis, while the advent of the World Wide Web, the digitization of

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publications, and later the rise of social media platforms opened the door to webometrics, full-text analysis, and altmetrics (Lindelöw et al., 2025).

Spectacular as these advancements have been, existing data sources still present several limitations when confronted with the “significant other” of science, the humanities (Franssen & Wouters, 2019). Citation indices were designed around the communication habits of the sciences, focusing on journals as the primary outlets for dissemination and framing citations as acknowledgments of intellectual debts (Garfield, 1955; Merton, 1988). In the humanities, however, publication practices are more diverse in terms of outlet types, target audiences, and languages (Knöchelmann, 2023), and citations often serve functions that go beyond straightforward attribution or cumulative knowledge building (Hellqvist, 2009).

These differences limit the extent to which citation data can be used as direct indicators of influence or intellectual credit in the humanities. At the same time, when used with appropriate caution and for analytically circumscribed purposes, citation relations can still provide meaningful relational information, such as patterns of co-attention, topical proximity, and shared problem spaces. In this sense, citation indices remain a useful resource for quantitative analyses—at least for those humanities fields that fall within their coverage.

More recent databases that provide full texts, such as JSTOR (Burns et al., 2009), have opened new opportunities for quantitative studies of the humanities. Nevertheless, they continue to suffer from limitations in data quality, disciplinary coverage, and temporal scope (Colavizza et al., 2023; Harzing & Alakangas, 2016; Laubichler et al., 2019). Finally, a limitation common both to citation indices and full texts database is that they track only formal authorship. Since authorship in the humanities is still dominated by single authorship, standard databases thus systematically underrepresent collaboration ties in these fields, hindering the effective investigation of their social structure (Cronin et al., 2004; Díaz-Faes & Bordons, 2017).

In light of the limitations of existing resources, an increasing number of projects in digital humanities, scientometrics, and related fields are focusing on designing and developing tailored data sources to support high-quality quantitative analyses of scholarly production in the humanities (see e.g., Colavizza et al., 2018). The key advantage of this approach is that the design of the data resource can be aligned with the specific characteristics of the research object and the analytical questions at stake (Colavizza et al., 2023).

A recent example of such a tailored data source is EDHIPHY (Enriched Data for the History of Philosophy, freely accessible at www.edhiphy.org), a database specifically designed for bibliometric research on twentieth-century philosophy (Petrovich et al., 2024). EDHIPHY covers approximately 23,000 articles published in 12 philosophy journals between 1890 and 1979. Its main feature is the tracking of *mention links* between publications and philosophers. These links can be considered the equivalent of citations, which, up until the 1960s, were scarcely used by philosophers, rendering classical citation analysis ineffective for studying the development of philosophy in the past century. EDHIPHY data have been used successfully to quantitatively investigate the rise of empirical logicism in American philosophy (Verhaegh et al., 2025).

In this paper, we present SOPHIS (the *Social Observatory of PHilosophy of Science*), a database developed to support quantitative research on contemporary philosophy of science. SOPHIS is openly available on Zenodo (Petrovich et al., 2025).

Unlike existing resources, SOPHIS is specifically designed to capture both the *intellectual* and the *social* dimension of philosophy of science. Alongside an intellectual classification of philosophy of science articles, it includes detailed data on the individuals and organizations that contribute, both formally and informally, to the production of the field—information not available in structured form in other data sources. In this way,

SOPHIS enables multidimensional, fine-grained analyses that are impossible with existing resources.

The paper is structured as follows. The next section explains how the design of SOPHIS is guided by a multi-layer model of knowledge systems and reviews existing quantitative literature on philosophy of science. We then describe the construction of SOPHIS, covering field delineation, data collection, processing and cleaning, enrichment, and graph-based data representation. Next, we demonstrate SOPHIS' potential by applying it to three research questions on the social structure of contemporary philosophy of science. Finally, we outline possible directions for future development.

Overall approach and related literature

Following (Lalli et al., 2020; Renn et al., 2016), we model philosophy of science as a system composed of three inter-related layers: the semantic, the semiotic, and the social layer.¹

The *semantic layer* comprises the epistemic contents specific to philosophy of science. It includes concepts (e.g., confirmation), philosophical theories (e.g., falsificationism), methodological approaches (e.g., integrated history and philosophy of science), positions in debates (e.g., realism vs. anti-realism about scientific entities), research programmes (e.g., Bayesian philosophy of science), sub-fields (e.g., philosophy of life sciences), and other forms of epistemic content. Handbooks and encyclopedic entries of philosophy of science typically aim to provide a comprehensive representation of this layer or part of it.

The *semiotic layer* encompasses the material supports of philosophy of science, both print and digital, such as books, edited collections, journals, conference proceedings, online resources, repositories, handbooks, syllabi, and teaching materials. These enable the circulation of epistemic content within the scholarly community and between philosophers of science and external audiences. Unlike the abstract entities of the semantic layer, entities in the semiotic layer have a material instantiation and spatiotemporal coordinates (e.g., a journal is founded in a specific year, books circulate in some academic libraries but not others, and so on).

Lastly, the *social layer* includes the individuals and organizations that produce, reproduce, and transmit the semiotic objects carrying epistemic content. It comprises professional philosophers of science, academic institutions such as universities and research centers, professional associations, as well as academic publishers and funding agencies that provide financial support for research in the field. Like the entities in the semiotic layer, those in the social layer also have a physical embodiment. The crucial difference, however, is that the latter possess agency (individual or collective) while the former do not. For this reason, we reserve the label *actor* for entities in the social layer.

In practice, the three layers are tightly intertwined and cannot exist independently of one another. Their analytical separation, however, is useful for capturing the state of quantitative research on philosophy of science and for understanding how SOPHIS represents the field.

The increasing digitalization of large collections of semiotic objects, such as journal articles and citation records, has made their quantitative analysis comparatively straightforward. Most existing quantitative studies of philosophy of science, indeed, rely on

¹ Unlike the original framework proposed by Lalli et al., we refer to *layers* rather than *networks*, in order to remain neutral on whether entities in all layers can always be modeled as networks.

digital versions of semiotic objects to infer properties and relationships within the *semantic* layer. For example, (Weingart, 2015) has applied bibliographic coupling to identify a cluster of authors working in integrated history and philosophy of science, while (Wray & Bornmann, 2015) have used Referenced Publication Years Spectroscopy to individuate key moments in the history of the field. Citation data also been employed to evaluate the extent to which philosophy of science influences scientific fields, and vice versa (Khelfaoui et al., 2021; McLevey et al., 2018; Plaisance et al., 2021a, 2021b). Petrovich and Viola (2024), for instance, analyzed citation flows between philosophy and neuroscience journals to investigate the nexus between the two domains, Pradeu et al., (2021) examined citation contexts to pinpoint the specific epistemic contributions of philosophers of science to the sciences, while Bonino et al., (2022) assessed the impact of formal methods in philosophy of science using reference analysis. Furthermore, the availability of full texts in electronic format through databases such as JSTOR has enabled studies that apply techniques such as topic modeling and text mining to reconstruct the intellectual trajectory of philosophy of science over the past century (Malaterre & Lareau, 2022; Malaterre et al., 2019, 2021) or to assess the role of methodologies such as case studies within the field (Mizrahi, 2020). More recently, Malaterre and Lareau (2024) have used topical profiles of authors in philosophy of science, extracted from the full-text of their articles, to reconstruct the development of intellectual communities in philosophy of science between 1930 and 2017. By contrast, studies explicitly targeting the *social* layer of philosophy of science remain comparatively scarce. Existing contributions rely either on survey data obtained through questionnaires or on indirect proxies derived from the intellectual layer such as co-citation (Weingart, 2015) or topic similarity (Malaterre & Lareau, 2024). Two studies based on questionnaire data are Unterhuber et al., (2014) and Plaisance et al., (2021a, 2021b). The former surveyed the state of philosophy of science in Germany between 1992 and 2012, reporting statistics on gender, academic positions, research foci, and the incidence of external funding. The latter investigated the attitudes and experiences of philosophers of science toward socially relevant philosophy of science, examining how these vary with social characteristics such as gender and career stage. Beyond philosophy of science narrowly construed, Contreras et al. (2022) have provided a large-scale analysis of philosophy PhD graduates from English-speaking PhD programs (based mainly in the United States), showing, among other things, how the anglophone academic philosophy is starkly divided between a core of highly-prestigious programs and a periphery of less prestigious programs (see also De Cruz, 2018). SOPHIS follows the approach of the first strand of studies but introduces a specific focus on the social layer, which remains largely uncharted. Like the first group of studies, its starting material is the semiotic layer of philosophy of science. Semiotic objects are then used both to infer features of epistemic contents (the semantic layer), as in most existing studies, and to reconstruct properties of individual and collective actors (the social layer), thereby addressing at the same time a gap in the current literature and in the data resources available.

As will be detailed in the next section, the primary source for SOPHIS is publications in philosophy of science journals. Publications constitute the main type of object in the semiotic layer of philosophical fields (Petrovich, 2024). Their components provide key information on both the semantic and the social layer. Specifically, titles, abstracts, keywords, and cited references/citations represent epistemic contents and relationships among them, as decades of science mapping studies have demonstrated (Börner, 2010; Chen, 2013; Petrovich, 2021b; Small, 2003; Thijs, 2019). As we will explain in the next sections, SOPHIS leverages such textual metadata and citation links to map the internal structure of the semantic layer of philosophy of science.

On the other hand, the bylines of authors and their affiliations point to individual and collective actors (i.e., organizations) in the social layer. Many scientometric studies have indeed relied primarily on authorship and affiliation metadata to map the social structure of scientific fields, most notably through co-authorship networks that reveal patterns of formal collaboration (Melin & Persson, 1996; Newman, 2001). Other actors and activities belonging to the social layer have been investigated using data on editorial board memberships (Baccini & Barabesi, 2011), graduate programs (Contreras Kallens et al., 2022), and funding agencies (Aagaard et al., 2021), as well as by relying on intellectual similarity measures as proxies for social proximity, as discussed above.

Acknowledgements analysis constitutes a particularly valuable complementary approach to these analyses, as it allows researchers to uncover informal collaboration patterns, sub-authorship practices, and the role of funders and other supporting actors that typically escape formal authorship metadata (Cronin, 1995; Cronin et al., 2004; Desrochers et al., 2017; Paul-Hus et al., 2017).

In order to include this important section of the social layer, SOPHIS incorporates, alongside formal authorship and affiliation data, the information contained in the acknowledgment sections of academic articles. In addition, it covers professional associations and recipients of major prizes in the field. In this way, SOPHIS aims to offer a richer and more comprehensive representation of the social actors underpinning the production of philosophy of science, by tracing four types of social processes: formal contribution and informal contribution (pertaining to individual actors), and organizational support and financial support for research (pertaining to collective actors such as universities and funding agencies).

Compared to available data sources for philosophy and philosophy of science, such as the *Philosopher's Index*, *PhilPapers*, and *PhilSci-Archive*, SOPHIS differs in three key respects.² First, its primary aim is to support quantitative research on contemporary philosophy of science, rather than provide a platform for archiving preprints (like *PhilSci-Archive*) or facilitating bibliographic search (like the *Philosopher's Index* and *PhilPapers*). Its open-access nature allows analysts unrestricted use of all its data, whereas large-scale data retrieval from the other resources is either prohibited or requires specific agreements with the maintaining institutions. Second, SOPHIS includes data not available in other resources, notably information extracted from acknowledgments about informal collaboration and funding. These data enable detailed analyses of the social layer of the field that are impossible with existing resources. Third, SOPHIS represents data in a graph-database structure (see next section), which facilitates advanced forms of network analysis. This design allows researchers to derive multiple kinds of networks (e.g., co-authorship, co-mention, funding networks) and to explore the interconnectedness of semantic, semiotic, and social dimensions in ways that other resources cannot support.

² The *Philosopher's Index* (<https://philindex.org/>) is a bibliographic database covering scholarly research in philosophy since 1940. Produced by the *Philosopher's Information Center*, it indexes journal articles, books, anthologies, and book reviews across all areas of philosophy. The database provides abstracts, controlled vocabulary subject indexing, and detailed metadata. It requires subscription.

PhilPapers (<https://philpapers.org/>) is a large-scale online index and bibliography of philosophy research, maintained by the *Center for Digital Philosophy* at Western University. It combines a comprehensive database of philosophical works with a community-driven curation system, where registered philosophers can edit and classify entries. *PhilPapers* also hosts *PhilArchive*, an open-access preprint repository.

The *Philosophy of Science Archive* (*PhilSci-Archive*, <https://philsci-archive.pitt.edu/>) is an open-access digital repository for preprints in philosophy of science. Established in 2001 and maintained by the *University of Pittsburgh Library System*, it promotes open science practices, serving as the main repository for current work in philosophy of science.

SOPHIS is close in spirit to the Internet Philosophy Ontology (InPhO, Buckner et al., 2011), an online infrastructure designed to support the quantitative study of philosophy by providing a dynamic, data-driven ontology of philosophical concepts, authors, and traditions. InPhO combines natural language processing applied to large philosophical corpora, including the *Stanford Encyclopedia of Philosophy* and PhilPapers, with expert curation and review. Like SOPHIS, InPhO is committed to open access and provides programmatic access to its data through APIs and downloadable archives.

Despite these shared goals, the two projects differ substantially in scope and analytical focus. InPhO is primarily oriented toward the construction of a taxonomy of philosophical concepts and thinkers, and thus places strong emphasis on the intellectual layer of philosophy. SOPHIS, by contrast, is designed to support the integrated analysis of semantic, social, and institutional dimensions of philosophy of science. As a result, SOPHIS places particular emphasis on affiliation data, funding information, informal collaboration, official roles in professional associations, which are currently outside the core scope of InPhO.

For these reasons, SOPHIS should be seen not as an alternative to InPhO, but as a complementary infrastructure, designed to address a different set of research questions and analytical needs within the quantitative study of philosophy.

Constructing SOPHIS

Field delineation

The first step in constructing SOPHIS was the *bibliographic delineation* of philosophy of science, i.e., the identification of a collection of publications—i.e., items in the semiotic layer—that offers a good representation of philosophy of science, both in terms of geographical scope and methodological diversity (Zitt et al., 2019).

Following previous studies (Malaterre et al., 2021; Wray, 2010), we adopted a journal-based strategy, defining philosophy of science in terms of publications appearing in a set of widely recognized leading disciplinary journals. In particular, we focused on the following eight journals: *Erkenntnis*, *Philosophy of Science (PS)*, *Synthese*, *Studies in History and Philosophy of Science (SHPS)*, *British Journal for the Philosophy of Science (BJPS)*, *Foundations of Science*, *Journal for General Philosophy of Science (JGPS)*, and *European Journal for Philosophy of Science (EJPS)*. This list includes both European (e.g., *EJPS*) and American (e.g., *PS*) journals, as well as theoretically oriented (e.g., *Erkenntnis*) and historically oriented (e.g., *SHPS*) outlets, providing a balanced representation of the field.

To focus on the most recent period of philosophy of science and to ensure comparability with previous studies of philosophy including acknowledgments data (Petrovich, 2021a), we considered publications from 2005 to 2019. Of the 8327 publications that appeared in the target journals over this 15-year period, we retained research articles ($n = 6826$), excluding book reviews, commentaries, editorials, and other minor document types. The focus on research articles is motivated by the fact that, on the one hand, they constitute the main epistemic contributions to the development of the field, and on the other, they are more likely to include acknowledgments than other publication types.

We are aware that a field delineation based exclusively on journal articles necessarily omits important contributions to philosophy of science published in books, edited volumes, and other formats. Extending SOPHIS to systematically include books is therefore a natural direction for future work, as we will explain in the section “[Future developments](#)”.

The present focus on journals is nonetheless theoretically and historically motivated. Compared to many other areas of philosophy, philosophy of science has relied to a remarkable extent on journals as key instruments of professional coordination and intellectual dissemination. Major philosophical movements have actively shaped the field through journal-based infrastructures: *Erkenntnis* functioned as the official outlet of logical positivism; *Philosophy of Science*, the journal of the Philosophy of Science Association, played a central role in structuring research agendas in the United States; and the *British Journal for the Philosophy of Science* fulfilled a similar function in the United Kingdom (Dewulf, 2021; Verhaegh, Forthcoming). More recently, a new professional association such as the European Philosophy of Science Association (EPSA) has launched its own journal, further attesting to the centrality of journals in the communication system of the field.

For these reasons, while a journal-based delineation does not exhaust the intellectual and social landscape of philosophy of science, it captures a particularly relevant segment of its social and intellectual organization.

Due to differences in publication rates, journals are unevenly represented in the corpus: *Synthese* and *PS* contribute the largest shares of articles (39.0% and 15.8%, respectively), while *JGPS* and *EJPS* contribute the smallest (3.6% and 3.4%).

Starting from this corpus, the development of SOPHIS proceeded in four main steps: data collection, data processing and cleaning, enrichment with additional information, and, finally, data representation using a graph database.

Data collection

Full metadata for the article corpus were retrieved from Web of Science by downloading the complete bibliographic records, which include the cited references of the target articles, from WoS web interface.³ The metadata were then supplemented with the acknowledgment texts, which were manually collected from the electronic full-text versions of the articles. We always used the published version rather than preprints, as the latter sometimes omit acknowledgments.

We considered as relevant only those acknowledgments appearing in a dedicated section of the article (e.g., a standalone acknowledgments section, or the first or last footnote/end-note), leaving aside brief mentions embedded in the main text or standard notes.⁴ Funding statements, when located in a separate section, were treated as part of the acknowledgment and merged with it.

A total of 78.8% of the articles in the corpus ($n = 5376$) include acknowledgments, indicating a high level of acknowledgment intensity, albeit lower than that observed in analytic philosophy (93.4%; see (Petrovich, 2024)). The average acknowledgment is 60 words long (st. dev. = 40, median = 52, min = 4, max = 391).

³ Retrieval date: 25/03/2024. Note that cited-reference data are not included in the released version of SOPHIS as cited-reference data indexed in Web of Science are proprietary to Clarivate Analytics and cannot be redistributed as part of an open-access dataset.

⁴ Since our focus is on individuals and funding organizations mentioned in the acknowledgments, excluding these minor acknowledgments does not result in data loss: individuals mentioned in them are almost always also included in the main acknowledgment text, while funding organizations are never mentioned in minor acknowledgments.

Data processing and cleaning

As explained in the introduction, SOPHIS is designed to track the social dimension of contemporary philosophy of science by identifying the individual and institutional actors that shape the field. Formal contributions and organizational support are captured through standard metadata—namely, authorship and affiliation data. Informal contributions and financial support, by contrast, are typically reported in the acknowledgment texts and must be extracted from them. In all cases, intensive data cleaning is required to transform raw data into high-quality, structured information.

In the following sections, we explain first how we processed the acknowledgment texts to extract individual mentions, then how we consolidated acknowledgments and authorship data, and finally, how we identified funders.

The names of individuals mentioned in the acknowledgments—hereafter *acknowledgees*—were extracted using Named Entity Recognition (NER), a Natural Language Processing technique that identifies and classifies named entities occurring in pieces of natural language into predefined categories. Specifically, we used the NER module of the Python library *spaCy* (<https://spacy.io>) to extract named entities from the corpus of acknowledgment texts. This yielded approximately 49,000 mentions of around 20,000 distinct named entities, of which around 10,000 were classified as persons.

The raw NER output was then manually corrected for misclassifications. False positives—i.e., entities incorrectly classified as persons ($n = 805$, e.g., “Alexander von Humboldt Stiftung”)—were excluded from the list of acknowledgees. Conversely, false negatives—i.e., persons not classified as such ($n = 298$, e.g., “Matteo Colombo”)—were added manually.

Next, the resulting list of names was consolidated to ensure that different mentions of the same individual were assigned a single unique identifier. Due to the informal nature of acknowledgments, indeed, diminutives (e.g., “Jon Kvanvig”) are often used alongside full names (e.g., “Jonathan Kvanvig”), resulting in multiple variants that must be merged to ensure accurate mention statistics. Consolidation was performed iteratively, combining automated processing with manual validation. Specifically, name variants were first parsed into first name and surname using a custom Python script capable of handling multiple first names and surnames with particles. Clusters of names sharing the same surname were then manually inspected to identify true variants. We adopted a conservative merging strategy, assigning the same unique identifier to different name variants only when we were reasonably confident that they referred to the same individual. After this consolidation process, the initial 9837 variants were reduced to a refined list of 8952 distinct acknowledgees (−9%).

A similar issue affected authorship data from Web of Science, where author names may appear in multiple variants. Following manual consolidation based on shared surnames, the number of unique authors was reduced from 4836 to 4380 (−9.4%). Finally, the lists of acknowledgees and authors were merged, resulting in a total of 10,876 unique *individual actors*, i.e., persons who appeared as authors and/or acknowledgees in the articles covered by SOPHIS.

As for the extraction of *funders*, an alternative strategy was preferred over NER, as experiments showed that the wide variety of ways in which funding agencies are mentioned makes it extremely difficult to identify them uniquely after extraction. Specifically, we developed a small web application⁵ that enabled manual entry of unique identifiers

⁵ <https://funding-ack-app.onrender.com>.

for the funding organizations mentioned in the acknowledgments, retrieving them via the Research Organization Registry (ROR, <https://ror.org/>) API. This approach ensures that SOPHIS funding data are interoperable with ROR.⁶ In cases where funders changed name or identifier between the acknowledgement and the present (e.g., the Flemish Agency for Innovation by Science and Technology, which was merged into Flanders Innovation & Entrepreneurship (VLAIO) in 2016), we used the most recent ROR identifier, relying on the successor relationships tracked in ROR or verified through online research. 29.2% ($n = 1990$) of research articles mention external funding (37% of articles including acknowledgments) and a total of 515 different funding organizations based in 51 countries are mentioned. This percentage is significantly higher than observed in analytic philosophy, where only 16.5% of articles are externally funded (Petrovich, 2024).

Data enrichment

Data on research articles and on actors, both individual and institutional, were further enriched to expand the analytic possibilities of SOPHIS of the semantic and social layers of philosophy of science.

Capturing the semantic layer

Philosophy of science's semantic layer is segmented into sub-areas, which include both the philosophy of specific disciplines (e.g., philosophy of biology) and various methodological approaches to the field (e.g., formal versus historical approaches) (Humphreys, 2014). However, standard Web of Science subject categories do not reach this level of granularity. To provide a fine-grained mapping of the semantic layer of philosophy of science, we therefore adopted a data-driven approach based on the processing of metadata that most directly reflect the epistemic content of the articles, i.e., their titles, abstracts, and cited references.⁷

In particular, we first generated the *bibliographic coupling network* (Kessler, 1963; Waltman & van Eck, 2014) of our corpus to find clusters of articles that are intellectually similar, either in terms of topics or methodologies. This method, based on the computation of the number of shared references between publication (i.e., bibliographic coupling similarity), is a well-established science mapping technique that has proven effective in capturing the intellectual structure of various fields, including philosophy (see, among others, Noichl, 2019; Petrovich & Buonomo, 2018; Weingart, 2015). Specifically, we used VOSviewer (van Eck & Waltman, 2010) to extract the bibliographic coupling network from our corpus and then applied its clustering algorithm (Waltman et al., 2010) to partition the network. In order to capture the structure of philosophy of science at different levels of granularity, we computed clustering solutions at three increasing values of the resolution parameter (1.0, 1.6, and 2.0), which resulted in 9, 16, and 21 clusters, respectively.⁸

⁶ In the very few cases where a funding organization mentioned in the text was not listed in ROR, we chose to omit it in order to preserve interoperability. Only active ROR records were used; inactive or withdrawn records were excluded.

⁷ Keywords were excluded because they were available for only 62.4% of the articles and, when present, tended to be of limited informational value.

⁸ 82(1.2% of the corpus) articles that did not include cited references or were isolated in the bibliographic coupling networks were excluded from classification.

In the second step, we labelled the obtained clusters using keywords automatically extracted from the titles and abstracts of the papers assigned to each cluster. Specifically, we applied two strategies to generate keywords. On the one hand, we used keyBERT, a keyword extraction module based on the BERT large language model.⁹ KeyBERT identifies terms that are semantically similar to the document using contextual embeddings from BERT, making it particularly effective for short texts such as titles and abstracts. On the other hand, we used classic TF-IDF scores, computed either from titles alone or from a combination of titles and abstracts. The two strategies produced three keyword lists for each cluster, which served as an objective basis for assigning synthetic labels (e.g., *Philosophy of biology*, *Decision theory*) to clusters. The combination of these keyword extraction strategies helped mitigate the subjectivity inherent in cluster labeling, a notorious challenge in science mapping and data-driven classification of scientific publications (Tijssen, 1993).

As noted above, clustering was performed at different resolution levels to reflect the stratified and nested nature of philosophy of science sub-specialties. An interesting outcome of this approach is that it reveals *parent–child relationships* between specialties across levels of granularity. Remarkably, some clusters remain stable across all resolution levels, i.e., they consistently include nearly the same set of publications regardless of the resolution parameter (e.g., the clusters *Epistemology* and *Philosophy of physics*). Others, by contrast, branch into subareas as resolution increases, with their publications distributed across new clusters at finer levels. For example, the clusters *Models in science* and *Science and values* emerge as child clusters of the broader *Integrated history and philosophy of science* cluster at higher resolution levels. Similarly, the clusters *Philosophy of mathematics* and *Metaphysics* stem from the broader *Logic and philosophy of language* cluster. A detailed mapping of these inter-cluster relationships is presented in the flow diagram and Sankey network included in the Supplementary Materials.

Characterizing social actors

As noted above, among the key actors of the social layer of philosophy of science there are *individuals*, who may be authors, acknowledgees, or both, and *organizations*, which may provide affiliation, funding, or both to individual actors. In order to allow for multi-dimensional and rich analyses of the social layer, both types of actors were supplemented in SOPHIS with further information.

For affiliation organizations, we used the Scopus database of organizations to identify each institution's standardized name in Scopus and to assign its corresponding city and country. For funding organizations, we relied on ROR metadata to assign a country, as this is the most relevant attribute for geographic analysis.¹⁰

Individual actors, by contrast, were enriched with affiliation data, bibliometric statistics, gender, awards received, and membership in professional associations, as explained in the following sections.

Affiliation data Individual actors were linked to their affiliation organizations (typically universities) using the Scopus API, queried via the Python package *Pybliometrics* (Rose & Kitchin, 2019). In this case, the advantage of using Scopus over Web of Science is that

⁹ <https://maartengr.github.io/KeyBERT/index.html>.

¹⁰ Further ROR metadata can be linked to SOPHIS using the unique identifier of the funder.

Scopus employs a standardized list of organizations, which helps to avoid issues of disambiguation and inconsistencies in address metadata (Baas et al., 2020).

Specifically, the affiliations of individuals who authored at least one article in our corpus ($n = 4380$) were retrieved directly from Scopus using the DOI of their article.

The affiliations of individuals mentioned only in the acknowledgments ($n = 6496$), by contrast, were retrieved using the following method. First, a query was built for the Scopus Author Search API¹¹ using the first name and surname recorded in SOPHIS. When Scopus returned only one author profile, the acknowledgee was identified with that profile, regardless of the research area of their publications. When multiple author profiles were returned, on the other hand, the acknowledgee was identified with the profile having the highest number of publications in Arts and Humanities.¹² This heuristic was motivated by the observed homophilic tendency of philosophers to acknowledge other philosophers rather than colleagues from other disciplines (Petrovich, 2024, Chapter 6). Although this approach may potentially produce some false matches—particularly in cases where the acknowledged individual is not a philosopher but a scientist with the same name—it was necessary in order to handle very common names, which can return dozens of profiles in Scopus.

As a result of this process, 68.9% of the acknowledgees were matched with at least one author profile in Scopus. For approximately 60% ($n = 3566$) of them, Scopus returned only one match; while for the remaining 40% ($n = 2595$), the above-mentioned disambiguation heuristic was applied to individuate the match across multiple profiles.

Once an author profile was identified, we selected one random publication by that author from the year in which they were acknowledged and retrieved his or her affiliation(s) listed in that publication. If no publication was available in the exact year, we searched for another published at most two years before or after the acknowledgment year. This strategy allowed us to approximate the acknowledgee's institutional affiliation at the time of mention, thus accounting for academic mobility.

As a result of this procedure, 79.9% of the 24,537 total mentions to individuals were successfully matched with an affiliation. The difference between the proportion of matches at the individual level (68.9%) and at the mention level (79.9%) indicates that affiliation data were more likely to be retrieved for individuals who were more frequently acknowledged. This is expected, since highly mentioned individuals are more likely to be recognized scholars whose work is indexed in Scopus.

Bibliometric statistics, gender, awards, and associations We retrieved from Scopus several other attributes for actors, including their bibliometric statistics (total number of publications, total number of citations, and h-index),¹³ their ORCID identifier, and their prevalent Scopus area—defined as the Scopus category in which they had published the largest number of articles.¹⁴

¹¹ <https://dev.elsevier.com/documentation/AuthorSearchAPI.wadl>.

¹² The identification of an article's research area is based on Scopus's disciplinary classification of journals: all articles published in a given journal inherit the journal's category. (see https://dev.elsevier.com/sc_author_search_tips.html).

¹³ At the time of retrieval.

¹⁴ The list of 27 Scopus macro-areas is available at https://dev.elsevier.com/sc_author_search_tips.html. As expected, given the focus of SOPHIS and our disambiguation heuristic, ARTS (Arts and Humanities) is the dominant area, accounting for 72.6% of identified actors.

Moreover, we used each actor's first name and main country of affiliation to automatically assign gender using the *genderize.io* service (<https://genderize.io/>). The main country was defined as the country of affiliation where the actor received the highest number of mentions: for instance, if Jan Sprenger received 40 mentions while affiliated with Tilburg University (The Netherlands) and 14 while affiliated with the University of Turin (Italy), his main country was defined as the Netherlands. This geographic information was crucial for improving the reliability of gender assignment, particularly for names associated with different genders in different countries (such as Andrea, which is predominantly male in Italy but female elsewhere).¹⁵ We highlight that this method, albeit effective, has important limitations. First, *genderize.io* operates on a binary conception of gender, excluding non-binary or gender-fluid identities. Second, coverage and accuracy vary across languages and regions, with lower reliability for uncommon names or non-Western naming conventions. Consequently, SOPHIS gender assignments should be interpreted cautiously and treated as approximations rather than definitive classifications.

Lastly, to cover professional associations, we manually retrieved from official websites and online sources the names of members of the governing bodies (e.g., presidents and other officials) of prominent philosophy of science associations. We considered the following six associations: the Philosophy of Science Association (PSA), the European Philosophy of Science Association (EPSA), the Committee for Integrated History and Philosophy of Science (&HPS), the Society for Philosophy of Science in Practice (SPSP), the Consortium for Socially Relevant Philosophy of/in Science and Engineering (SRPoiSE), and the Society for the Metaphysics of Science (SMS).

Similarly, we retrieved the names of recipients of three major philosophy of science awards—the Lakatos Award, the Popper Prize, and the Hempel Award—and linked them to their corresponding SOPHIS profiles.

Data representation

SOPHIS data can naturally be conceived as a *network* of entities of different types, connected by various kinds of links. For example, actor-type entities can be connected to article-type entities via authorship or acknowledgment links, while article-type entities can be connected to subdiscipline-type entities via clustering links.

To reflect this underlying network structure, we organized the data in a *graph database* composed of nodes and edges, rather than in a traditional relational database composed of tables and relationships. A key advantage of graph databases over relational databases is that they facilitate the generation of derived networks and support the computation of network analytics, such as centrality measures (Robinson et al., 2015). The graph representation of SOPHIS data, for example, allows analysts to readily compute the co-authorship network or the co-mention network—where acknowledgees are linked by the number of publications in which they are jointly acknowledged (Petrovich, 2022)—as well as several other types of network projections (Batagelj & Cerinšek, 2013).¹⁶ SOPHIS was implemented in Neo4j, one of the most widely used graph database systems (<https://neo4j.com/>).

¹⁵ Note that the main country may not coincide with the nationality of an actor. In this case, for instance, Jan Sprenger is German, not Dutch.

¹⁶ Note that, in addition to the graph-based version, we also provide a standard relational version of SOPHIS. Its schema diagram is included in the Appendix. The Zenodo repository contains both the tables in.csv format and the relational database in SQLite format.

Table 1 Properties of node types in SOPHIS

Node type (abs. and rel. frequency)	Description	Properties
ACTOR (10,876, 53.4%)	Represents an individual actor who either authors an article or is mentioned in the acknowledgments of an article	<i>H_ID</i> (Unique SOPHIS identifier of the actor) <i>Standard_Label</i> (Standard format of the actor's name in SOPHIS) <i>First_Name</i> ^N (First name, as parsed by the algorithm and manually validated) <i>Surname</i> (Surname, as parsed by the algorithm and manually validated) <i>Gender</i> ^N (Gender attributed by genderize.io, values = <i>male</i> , <i>female</i>) <i>Gender_Probability</i> ^N (Probability of the gender attribution) <i>Author_Scopus_ID</i> ^N (Unique identifier on Scopus) <i>ORCID</i> ^N (ORCID of the actor) <i>Citations</i> ^N (Number of citations in Scopus at retrieval date) <i>H_Index</i> ^N (H-index recorded in Scopus at retrieval date) <i>Scopus_Area</i> ^N (Main Scopus area of the actor's publications) <i>Tot_Publications</i> ^N (Number of publications in Scopus at retrieval date) <i>UT</i> (Unique WoS identifier of the article) <i>DOI</i> ^N (DOI of the article) <i>EID</i> ^N (Unique Scopus identifier of the article) <i>Title</i> (Title of the article) <i>Pub_Year</i> (Publication year) <i>Journal_Complete</i> (Journal name in WoS) <i>Journal_Abb</i> (Journal name abbreviation in WoS) <i>Keywords</i> ^N (Author keywords in WoS) <i>Abstract</i> ^N (Abstract of the article) <i>Volume</i> ^N (Volume of the article) <i>Issue</i> ^N (Issue of the article) <i>Beginning_Page</i> (Beginning page of the article) <i>Ending_Page</i> (Ending page of the article) <i>Ack_Text</i> ^N (Text of the acknowledgment, manually collected from the article)
		<i>Inst_ID</i> (Unique SOPHIS identifier of the institution) <i>Affiliation_Name</i> (Standard name of the institution in Scopus) <i>Affiliation_City</i> ^N (Standard name of the city of the institution in Scopus) <i>Affiliation_Country</i> (Standard name of the country of the institution in Scopus) <i>Affiliation_Country_ISO</i> (ISO 3166-1 alpha-2 code of the country of the institution)
ARTICLE (6826, 33.5%)	Represents an article covered by SOPHIS	
INSTITUTION (2105, 10.3%)	Represents an institution the actors are affiliated with	

Table 1 (continued)

Node type (abs. and rel. frequency)	Description	Properties
FUNDER (515, 2.5%)	Represents a funding body mentioned in the acknowledgements	<i>Funder_ID</i> (Unique ROR identifier of the funder) <i>Funder_Label</i> (Standard name of the funder in ROR) <i>Funder_Country</i> (Country of the funder in ROR) <i>Funder_Country_ISO</i> (ISO 3166-1 alpha-2 code of the country of the funder) <i>Cluster-nodes can exist at different levels of resolution (i.e., one cluster can exist in one, two or all three levels of resolution). Then, for each level of resolution, the node includes the following properties:</i> <i>Coarse</i> (1 if the cluster exists at coarse resolution level, 0 otherwise) <i>Intermediate</i> (1 if the cluster exists at coarse resolution level, 0 otherwise) <i>Fine</i> (1 if the cluster exists at coarse resolution level, 0 otherwise) <i>Num_cluster</i> (Unique identifier of the cluster at the given level of resolution) <i>Keywords_keyBERT</i> (Cluster keywords extracted with keyBert from a combination of titles and abstracts of articles in the cluster) <i>Keywords_tf_idf_AB</i> (Cluster keywords based on tf-idf applied to abstracts of articles in the cluster) <i>Keywords_tf_idf_TI</i> (Cluster keywords based on tf-idf applied to titles and abstracts of articles in the cluster) <i>Cluster_label</i> (Summary label of the cluster, human-assigned) <i>Name_association</i> (Official acronym of the association)
CLUSTER (23, 0.1%)	Represents a sub-discipline resulting from the clustering of articles based on bibliographic coupling similarity	
ASSOCIATION (7, <0.1%)	Represents the association in which actors play governing roles	
PRIZE (3, <0.01%)	Represents a philosophy of science prize	<i>Prize</i> (Name of the prize)

^NCan be NULL

Table 2 Properties of link types in SOPHIS

Link type (abs. and rel. frequency)	Description	Direction	Multiplicity	Properties
MENTIONS (49,072, 33.3%)	Occurs when an article mentions an entity in its acknowledgments	(:article))-[:MENTIONS]-> (:actor)	No	<i>Ent_Category</i> (Category assigned to the entity by spaCy. Note: all extracted entities are included, but only those classified as <i>PERSON</i> generate links with actor-nodes) <i>Ent_ID</i> (SOPHIS unique identifier of the entity) <i>Ent_Text</i> (Original string extracted by spaCy) <i>Mention_ID</i> (Identifier of the mention relationship in the relational version of SOPHIS)
IS_CLUSTERED_IN (39,410, 26.7%)	Occurs when an article is assigned to a cluster	(:article))-[:IS_CLUSTERED_IN]-> (:cluster)	Yes	<i>Resolution</i> (Resolution level of clustering; one link per article—cluster assignment per resolution)
IS_AFFILIATED_WITH (38,142, 25.9%)	Occurs when an actor is affiliated with an institution	(:actor))-[:IS_AFFILIATED_WITH]-> (:institution)	Yes	<i>Year</i> (Affiliation year; one link per year of actor—institution affiliation) <i>Aff_ID</i> (Identifier of the affiliation relationship in the relational version of SOPHIS)
IS_AUTHOR_OF (17,146, 11.6%)	Occurs when an actor (co-) authors an article	(:actor))-[:IS_AUTHOR_OF]-> (:article)	No	<i>Au_Number</i> (Position of the author in the byline) <i>Indexed_Name_Scopus</i> (Author name as indexed in Scopus) <i>WoS_Author</i> (Author name as indexed in WoS) <i>AT_ID</i> (Identifier of the authorship relationship in the relational version of SOPHIS)
FUNDING (2,807, 1.9%)	Occurs when a funding body funds an article	(:funder))-[:FUNDING]-> (:article)	No	
IS_ASSOCIATED_WITH (832, 0.6%)	Occurs when an actor plays a governing role in an association	(:actor))-[:IS_ASSOCIATED_WITH]-> (:association)	Yes	<i>Role</i> (Role held by actor in the association; one link per actor-association role)
HAS_WON (128, 0.1%)	Occurs when an actor wins a prize	(:actor))-[:HAS_WON]-> (:prize)	Yes	<i>Award_Year</i> (Year prize awarded; one link per actor-prize award)

SOPHIS currently includes 20,355 nodes and 147,537 links. Nodes are distributed across seven types (i.e., modes, in network-analytic terms), and there are also seven types of links. Tables 1 and 2 report, respectively, the properties of node types and link types. Figure 1 presents the SOPHIS meta-graph, showing the possible relationships between node types.

Actor-nodes can be connected with four different node types: prize-type, when they have received an award, association-type, when they hold a position in a professional association, institution-type, when they are affiliated with an academic institution, and article-type, when they author of one or more articles. Actor-nodes and article-nodes can also be linked in the opposite direction when an actor is mentioned in the acknowledgments of one or more articles. Article-nodes, in turn, can be connected to two additional node types: funder-type, when acknowledgements mention one or more funding bodies, and cluster-type when articles are assigned to a cluster. From these primary relationships, it is possible to generate second-order relationships between node types through suitable network projections. For instance, co-authorship links between actor nodes can be derived from the primary relationship `IS_AUTHOR_OF` between actor nodes and article nodes.

SOPHIS adopts a multigraph structure, allowing multiple links between specific pairs of nodes when required by the data. For example, an actor may be connected to the same institution by multiple links if affiliated with that institution in different years, or to the same professional association when holding multiple roles. By contrast, multiple links between actor-nodes and article-nodes are not allowed: an actor cannot have more than one authorship relationship with the same article and cannot be acknowledged more than once in the same article. Similarly, multiple links cannot exist between a single funder and a single article.

The degree centrality of nodes in SOPHIS can be straightforwardly interpreted when broken down by source or target node type. For instance, the out-degree centrality of actor nodes, when disaggregated by the four target node types, corresponds respectively to the number of prizes received, the number of affiliated associations, the number of affiliated institutions, and the number of (co-)authored articles.

Links in SOPHIS are also characterized by properties that qualify the relationship between nodes and enable the application of suitable filters when needed. For example, the link `IS_AFFILIATED_WITH` has a temporal property (*Year*), indicating the year in which an actor was affiliated with a given institution. Thanks to this temporal tag, SOPHIS can efficiently represent actor mobility across institutions over time. Similarly, the link `IS_CLUSTERIZED_IN`, which associates an article with a cluster, has a property (*Resolution*) that specifies the granularity level of the clustering solution. This enables the analyst to select the desired level of granularity. Finally, the link `IS_ASSOCIATED_WITH` includes a property (*Role*) that specifies the governing role played by the actor in an association (e.g., President, Committee Member). Detailed descriptions of link properties are provided in Table 2.

SOPHIS in action

In this section, we demonstrate the potential of SOPHIS by presenting a set of illustrative analyses that showcase the kinds of quantitative insights the database enables into the social structure of contemporary philosophy of science. These analyses are intended as proof-of-concept examples rather than as exhaustive or definitive characterizations of the field. All results are necessarily relative to the specific corpus underlying SOPHIS (eight core journals in philosophy of science and a contemporary time window spanning 2005–2019)

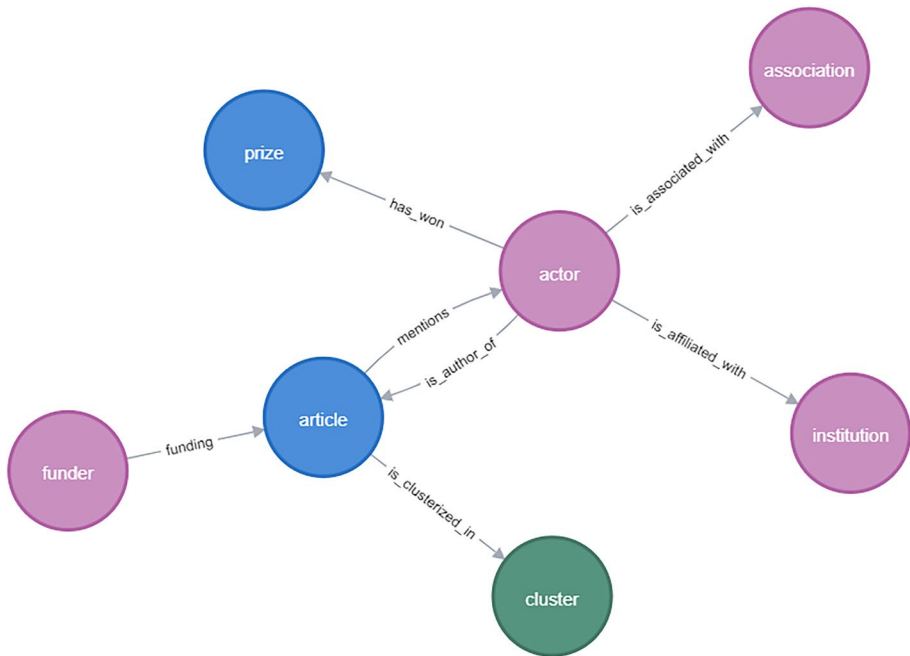


Fig. 1 SOPHIS meta-graph. Entities of the social layer in purple, of the semiotic layer in light blue, of the semantic layer in green

and should be interpreted accordingly. Within these constraints, we use SOPHIS to explore three broad questions that exemplify its analytical affordances:

- Which actors—individuals, institutions, or countries—contribute most to the production of philosophy of science, both formally and informally? Is prestige concentrated among a few actors, or is it more evenly distributed?
- What characteristics of individual actors most strongly influence their prestige within the field?
- How is the social layer of philosophy of science structured? Do communities of philosophers of science organize primarily around topics, methodological approaches, or professional associations?

These questions all relate to the topic of *prestige* in philosophy of science. In the sociology of science, prestige is considered a key resource that scientific actors aim to accumulate (Merton, 1974). Individuals and institutions endowed with high prestige accumulate various forms of academic capital and can convert one form of capital into another (Bourdieu, 2008) more easily than less prestigious actors, enabling them to shape the social and epistemic dynamics of their disciplines. Although essentially an immaterial property, prestige is manifested through institutional devices such as prizes, chairs, and reputation rankings.

In scientometrics, the theory of the “reward triangle” distinguishes three indicators of prestige related to the scientific communication process: authorship, citations, and

Table 3 Most productive authors

Author	Gender	Articles	Association(s)
Gerhard Schurz	M	22	EPSA
Igor Douven	M	18	
Jeff Barrett	M	17	PSA
Roman Frigg	M	17	EPSA
Jan Sprenger	M	17	EPSA
Darrell Rowbottom	M	17	
Juha Saatsi	M	16	
William Bechtel	M	16	
Mikhail Katz	M	16	
John Norton	M	15	&HPS, PSA

Authors with at least 15 articles in SOPHIS. None of them received an award

acknowledgments (Costas & Leeuwen, 2012; see also Cronin & Weaver-Wozniak, 1995).¹⁷ Compared to authorship and citations, acknowledgments have been less studied and theorized, partly because they are not routinely collected by standard databases. However, they are especially valuable in fields such as philosophy of science, where authorship captures only part of the social networks underlying publications and therefore does not fully reflect contributions.

It is important to remember, however, that using acknowledgments as proxies of prestige has limitations. Like authorship and citations, acknowledgments can be subject to strategic behavior: for a discussion on the two theoretical interpretations of acknowledgements, the normative and the strategic, we refer to (Baccini & Petrovich, 2022). Moreover, acknowledgments do not capture other dimensions of prestige, such as editorial power or gatekeeping capacity. Thus, they represent one among several complementary measures of prestige rather than a complete characterization of social influence in the field.

Top actors and distribution of prestige

To address the first research question, we draw on two types of SOPHIS data: *authorship*, which captures the formal contribution of actors through published work, and *mentions in acknowledgments*, which we use as a proxy for informal contributions such as peer interaction, mentoring, and other forms of support (Cronin et al., 2003; McCain, 2024).

Tables 3 and 4 report, respectively, the most productive authors and the most frequently acknowledged individuals, together with information on their gender, governing roles in professional associations, and awards received. A noteworthy pattern emerges: presidents of professional associations and recipients of awards occupy the highest ranks in acknowledgments, though not in authorship. For instance, Elliott Sober (rank 1) and James Woodward (rank 8) both served as presidents of the PSA (2003–2004 and 2011–2012, respectively), while Stephan Hartmann (rank 5) was president of EPSA (2013–2015). These cases

¹⁷ In the “science of science”, prestige is often measured by computing centrality measures over networks of scientific actors (Fortunato et al., 2018; Newman, 2018). For an application to philosophy, see Contreras Kallens et al. (2022), who use eigenvector centrality in the academic placement network of PhD graduates as a measure of prestige.

illustrate the capacity of these actors to shape the field by both institutional means (governing professional associations) and by informal support and advice to other philosophers of science. Notably, both Sober and Woodward have also received prestigious prizes, suggesting that acknowledgment mentions correlate with the formal recognition of prestige.

An additional interesting case is David Chalmers, a prominent analytic philosopher rather than a philosopher of science *stricto sensu*, as it is confirmed by the fact that he has not authored any paper in the journals covered by SOPHIS and by the fact that more of 80% of his mentions come from *Synthese* and *Erkenntnis*, the two most analytically-oriented journals in the corpus.¹⁸

Both rankings reveal a pronounced gender imbalance: men consistently occupy the top positions. The most productive woman, Charlotte Werndl, has authored 13 articles, while the most acknowledged woman, Nancy Cartwright (recipient of the Hempel Award in 2018), has 53 mentions.

Looking beyond the top positions, the Pearson correlation between productivity and acknowledgments of all individual actors is 0.52, indicating a moderate association between formal and informal contributions.

Turning to institutional actors, Tables 5 and 6 report the top five most influential universities and countries. For each institution and country, we calculated the number of mentions, authorships, distinct authors, and officials. Mentions of an institution are defined as the sum of all mentions received by actors affiliated with that institution at the time of acknowledgment. Authorships are computed in the same way, counting only the publications produced while authors were affiliated with the institution.¹⁹

All four measures of influence are strongly correlated at both institutional and national levels. For institutions, mentions, authorships, and distinct authors correlate above 0.90, while the number of officials correlates at approximately 0.70 with the other three. For countries, all four variables correlate above 0.96.

The list of top institutions includes universities that host world-leading departments or research centers in philosophy of science. The first in the ranking, the University of Pittsburgh, hosts the Center for Philosophy of Science, founded in 1960 by Adolf Grünbaum, a key figure in the field during the second half of the Twentieth century. The London School of Economics and Political Science (rank 2) was the academic home of Karl Popper and his school, while the University of Munich (rank 3) hosts the Munich Center for Mathematical Philosophy (MCMP), a major hub for formal methods in philosophy of science. The University of Oxford (rank 4), one of the world's largest centers for analytic philosophy, is also institutionally linked to the *British Journal for the Philosophy of Science*. Finally, the Department of History and Philosophy of Science of the University of Cambridge (rank 5) is a key center for integrated history and philosophy of science approaches.

At the country level, English-speaking countries dominate, with the United States, United Kingdom, and Canada ranking first, second, and fifth, respectively. Germany and the Netherlands complete the top five, and the strong presence of the Netherlands is particularly noteworthy given its relatively small size. By contrast, the leading position of the United States is less surprising, reflecting both the country's vast academic system and its central role in global science. Interestingly, these five countries also fund approximately

¹⁸ Chalmers is by large the most mentioned philosophers in analytic philosophy journals, being mentioned in approximately 1 out of 20 articles appeared in the most prestigious analytic philosophy journals in 2005–2019 (Petrovich, 2024, Chapter 6).

¹⁹ In cases of multiple affiliation, no fractionalization is applied: the entire mention or authorship is attributed to each listed institution.

Table 4 Most mentioned acknowledgees

Acknowledgee	Gender	Mentions	Association(s)	Award(s)
Elliott Sober	M	104	PSA*	Lakatos, Hempel
John Norton	M	97	&HPS, PSA	
Carl Craver	M	66		
Anjan Chakravartty	M	65	SRPoiSE, PSA	
Stephan Hartmann	M	62	EPSA*, PSA	
David Chalmers	M	61		
Alan Hajek	M	60	PSA	
James Woodward	M	57	PSA*	Lakatos
Branden Fitelson	M	55		
Jeremy Butterfield	M	54		
Hannes Leitgeb	M	54		

Top ten most mentioned actors

*Served as president of the association

19% of the articles covered by SOPHIS, highlighting their key role as supporters of philosophy of science research.

In addition to focusing on the most influential actors, it is instructive to examine the overall distribution of prestige in contemporary philosophy of science, to assess whether it exhibits the same concentration dynamics observed in science more generally (Bol et al., 2018; Merton, 1988; Seglen, 1992). The two panels in Fig. 2 show the Lorenz curves for mentions and authorships, respectively, broken down by individuals and institutions.

In both cases, we observe substantial inequality typical of skewed distributions: a small number of actors—whether individuals or institutions—account for a large share of mentions or authorships, while the vast majority contribute relatively little. For example, the top 10% of most productive authors account for 35% of authorships, whereas the top 10% of most mentioned acknowledgees capture 50% of all mentions. At the institutional level, the top 10% of institutions account for 60% of authorships and approximately 75% of mentions.

Gini coefficients are consistently higher for institutions than for individuals (0.686 vs. 0.385 for authorships; 0.8 vs. 0.535 for mentions), and inequality is more pronounced for mentions than for authorships. These results indicate that prestige, while already concentrated at the individual level, is even more sharply concentrated at the institutional level, supporting the view that the institutional structure of contemporary philosophy of

Table 5 Most influential institutions

Institution (city, country)	Mentions	Authorships	Authors	Officials
University of Pittsburgh (Pittsburgh, USA)	374	171	78	17
London School of Economics and Political Science (London, UK)	270	158	55	8
Ludwig-Maximilians-Universität München (Munich, Germany)	186	132	62	4
University of Oxford (Oxford, UK)	228	106	61	2
University of Cambridge (Cambridge, UK)	263	103	57	5

Top 5 institutions by number of mentions are shown

Table 6 Most influential countries

Country	Mentions	Authorships	Authors	Officials
United States	2955	3020	1589	104
United Kingdom	1843	1339	674	37
Germany	814	699	389	15
The Netherlands	492	481	238	14
Canada	674	397	243	16

science—dominated by a relatively small number of elite institutions—amplifies inequalities among individuals.

These patterns resonate with the findings of Contreras et al. (2022), who similarly observe a stark divide between high- and low-prestige graduate programs in anglophone philosophy. In both analyses, a relatively small number of actors—whether individuals, institutions, or programs—capture a disproportionate share of recognition or influence, highlighting the highly unequal distribution of prestige within the field. While our study focuses on acknowledgements and authorships, and theirs on placement dynamics in academic philosophy (not only philosophy of science), the convergence in concentration patterns underscores that both approaches reveal the central role of elite actors and institutions in shaping the social landscape of academic philosophy.

Factors affecting individual prestige

In this second study, we aim to explore how the facet of prestige reflected in acknowledgments is associated with the personal characteristics of individual actors in the philosophy of science community. Future research could further examine how other dimensions of prestige and symbolic capital relate to, or interact with, acknowledgment-based recognition.

Specifically, we apply a negative binomial regression model to estimate the effect of various individual-level variables on mentions. The choice of this model is motivated by the skewed distribution observed in the previous section, as the negative binomial appropriately handles overdispersion in count data (see e.g., Rao, 1980).

The explanatory variables include: number of awards received, number of publications in philosophy of science journals, number of governing roles in professional associations, affiliation with at least one English-speaking country during the career, gender, total number of publications, total number of citations, and number of externally funded articles. Together, these capture different dimensions of academic standing—further proxies of prestige (awards, association roles, citations), intra-field and overall productivity, institutional background, personal characteristics (gender), and capacity to attract funding. Descriptive statistics of the variables included in the model are reported in Table 7.

Only actors with complete data across all variables were included, yielding a sample of 6477 observations (59.6% of all actors in SOPHIS). Importantly, this sample also includes actors with 0 mentions—that is, authors never mentioned in the acknowledgments.

Mean, 1st quartile, median, 3rd quartile, minimum and maximum values are reported for quantitative variables; relative frequencies are reported for categorical variables.

Table 8 reports the results of the regression. Formal recognition through prizes and institutional background are among the strongest drivers of mentions. Awards stand out as

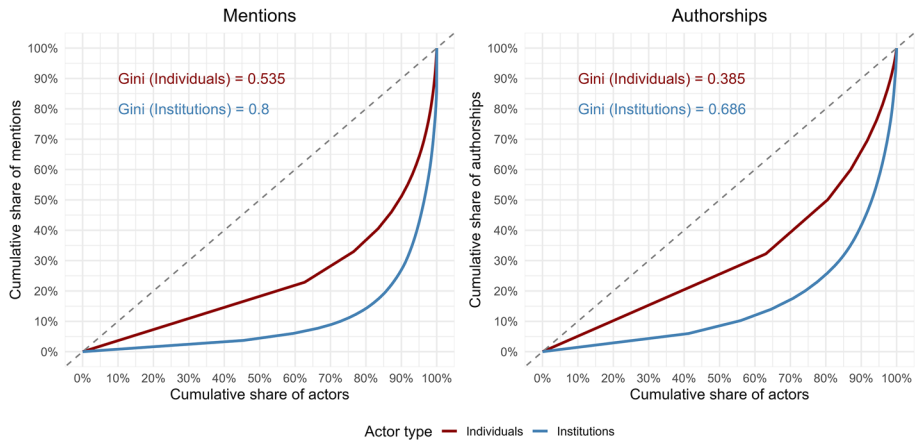


Fig. 2 Lorenz curves for mentions (left panel) and authorships (right panel). The horizontal axis reports the cumulative proportion of actors (ranked from least to most mentioned/authored), while the vertical axis reports the cumulative proportion of total mentions or authorships they account for. The diagonal line represents perfect equality. Deviations from the diagonal indicate increasing concentration, with more strongly bowed curves corresponding to higher inequality in the distribution

the most powerful predictor: each additional award is associated with a 144% increase in expected mentions. Similarly, affiliation with an English-speaking country raises expected mentions by 91%, confirming the dominance of the English-speaking world in philosophy of science observed above. Interestingly, productivity matters only within the field: each philosophy of science publication adds approximately 28% more mentions, while sheer productivity has only a negligible effect (circa 0.23% per paper). Similarly, the overall citation count obtained in Scopus has almost no effect on mentions, suggesting that prestige in philosophy of science is driven by recognition within the field rather than visibility in the overall scientific literature (see also McLevey et al., 2018). Serving in professional associations also boosts mentions by circa 23%, confirming that governing roles provide opportunities to shape the field both formally and informally. Gender exerts a moderate influence, with men expected to receive approximately 17% more mentions than women, consistent with broader patterns of gender disparities in recognition in science and other areas of philosophy (Petrovich, 2024). Men are indeed disproportionately represented in the population, accounting for 78.8% of authors and 77.4% of acknowledgees. Finally, the number of funded articles shows a small and non-significant negative effect, suggesting that funding, at least in this context, does not straightforwardly translate into greater influence.

Mapping the social layer of philosophy of science

As noted above, the graph-based design of SOPHIS enables the construction of second-order networks that provide valuable insights into the social structure of contemporary philosophy of science. In this final study, we leverage this feature to map the field's fine-grained social organization and identify its main communities.

In philosophy of science, as in other areas of philosophy and the humanities more broadly, formal collaboration in the form of co-authorship is relatively rare. For instance, in SOPHIS, 56% of authors have no co-author and 22% have only 1 co-author, with 81%

Table 7 Descriptive statistics of variables included in the model

Response variable	Description	Mean	1st Qu	Median	3rd Qu	Min	Max
<i>Mentions</i>	Number of mentions received in the whole timespan covered by SOPHIS	2.49	0	1	2	0	104
Quantitative explanatory variables	Description	Mean	1st Qu	Median	3rd Qu	Min	Max
<i>Awards</i>	Number of awards received	0.01	0	0	0	0	2
<i>PhilSciPubs</i>	Number of publications in journals covered by SOPHIS	1.07	0	1	1	0	22
<i>GovRoles</i>	Number of governing roles in professional associations	0.06	0	0	0	0	14
<i>Pubs</i>	Total number of publications (in Scopus)	39.07	10	21	43	1	1781
<i>Cits</i>	Total number of citations (in Scopus)	1347	48	186	650	0	228,493
<i>Funding</i>	Number of externally funded articles (co-)authored	0.36	0	0	0	0	12
Categorical explanatory variables	Description	Frequency					
<i>Eng</i>	Affiliation with at least one English-speaking country during the timespan covered by SOPHIS	Yes (= 1): 0.59 No: 0.41					
<i>Gender</i>	Gender assigned by <i>genderize.io</i> . Observations without gender are removed	Male (= 1): 0.78 Female: 0.22					

Table 8 Estimated parameters (percentage change in parentheses), Standard Errors (S.E.) and Significance (Sig.) values for the model

Variable	Estimate (% change)	S.E	Sig
Intercept	−0.221 (−19.855%)	0.039	***
<i>Awards</i>	0.889 (+143.485%)	0.124	***
<i>PhilSciPubs</i>	0.249 (+28.27%)	0.011	***
<i>GovRoles</i>	0.209 (+23.342%)	0.027	***
<i>Pubs</i>	0.002 (+0.233%)	< 0.001	***
<i>Cits</i>	−0.00002 (−0.002%)	< 0.001	***
<i>Funding</i>	−0.003 (−0.312%)	0.020	
<i>Eng</i>	0.649 (+91.442%)	0.031	***
<i>Gender</i>	0.159 (+17.298%)	0.037	***

***Significant at 0.1%

of articles being mono-authored. Malaterre and Lareau (2024) report a share of 16% co-authored papers in philosophy of science in the period 1996–2017. In cases like this, co-authorship networks are too sparse to be informative of social structures. Informal collaborations recorded in acknowledgments, hence, become crucial. In SOPHIS, 68% of the articles acknowledge at least one person, and articles featuring acknowledgments mention on average 4.5 individuals (st. dev.=4.4, median=3). These figures suggest that informal collaboration data are dense enough to compensate for the limitations of formal collaboration data.

To map the social layer of philosophy of science, thus, we adopted the technique of *co-acknowledgment networks*, where two actors are linked when they are mentioned together in the same acknowledgments and the strength of the link corresponds to the number of acknowledgments they have in common. This approach has proven effective in other areas of philosophy for uncovering underlying communities of actors (Petrovich, 2024). For clarity of visualization and interpretation, we restricted the network to addressees with at least 10 mentions in SOPHIS ($n = 447$, corresponding to the top 5% of all addressees). The resulting map, produced with VOSviewer, is presented in Fig. 3.

In the visualization, each node represents an actor, with node size proportional to the number of mentions received. The distance between nodes reflects co-acknowledgment similarity: actors frequently mentioned together are placed closer, while those rarely mentioned together are farther apart. Node colors indicate clusters as identified by the VOSviewer clustering algorithm, which groups actors who are frequently mentioned together in the same acknowledgments.

The assignment of labels to these clusters proceeds in a separate step and relies on the article-level classification obtained through bibliographic coupling. As described above, bibliographic coupling is used to assign each article in the corpus to a research area (sub-domain) of philosophy of science. For each actor, we then examined the distribution of

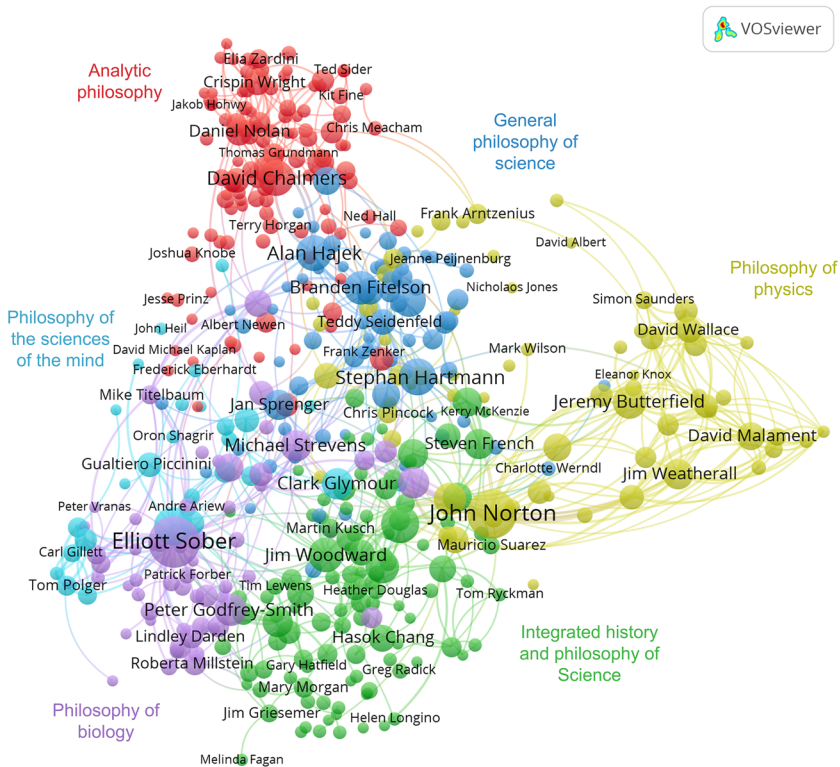


Fig. 3 Map of the social layer of philosophy of science. Nodes represent actors receiving at least 10 mentions in SOPHIS. Links indicate the number of co-acknowledgments (i.e., mentions within the same acknowledgment section) between actors; only links with a strength of at least 4 co-acknowledgments are shown. Node size is proportional to the total number of mentions received. Distances between nodes reflect co-acknowledgment similarity: actors frequently mentioned together are positioned closer to each other, while actors rarely mentioned together are placed farther apart. Node colors indicate clusters identified using the VOSviewer clustering algorithm (resolution parameter=0.9). An interactive version of the map, with further data and statistics, is available at <https://tinyurl.com/2bshcfkw>

these research areas across the articles in which the actor is acknowledged. When an actor is also an author in the corpus, we additionally considered the research areas of the articles authored by that actor.

These actor-level profiles were accessed through the online interactive version of the map (available at <https://tinyurl.com/2bshcfkw>), which provides summary statistics for each node.²⁰

Cluster labels were then assigned manually by inspecting the aggregate profiles of the actors within each cluster. For example, Hasok Chang received a total of 39 acknowledgments, of which 41% originated from articles classified as Integrated History and

²⁰ The information includes: gender, main affiliation, governing roles in associations, award(s) received, total mentions, with breakdown by sub-area, total publications, with breakdown by sub-area, cosine similarity between vectors of mentions by area and articles by area, Spearman similarity between the same two vectors, titles of publications (linked to articles via DOI), keywords associated with articles in WoS, cluster assignment (human-assigned label). Gender and associations can be visualized on the map by changing the color property of the nodes.

Philosophy of Science, 13% from Philosophy of Biology, and smaller shares from Causation, Models in Science, Science and Values, Scientific Realism, and Mechanisms. His authored publications in the corpus display a similar profile, with a strong concentration in Integrated History and Philosophy of Science. Since most actors in the same cluster exhibit comparable distributions, Cluster 2 was collectively labeled “Integrated History and Philosophy of Science”. Cluster labels should be interpreted with caution. They are intended as heuristic devices to support the interpretation of the map, rather than as definitive classificatory categories for individual philosophers of science.

Subject to this caveat, the map reveals that the social layer of contemporary philosophy of science is structured around both methodological approaches and substantive research themes. Along the vertical axis, we observe at one end Cluster 1, which gathers mainly analytic philosophers rather than philosophers of science, and at the other end Cluster 2, which includes philosophers of science who privilege the use of historical case studies in their work. In between lies Cluster 3, composed of philosophers of science focusing on general epistemological issues (e.g., theory, confirmation) and frequently employing formal methods from probability theory. By contrast, the horizontal axis highlights thematic rather than methodological clusters. Cluster 6 brings together philosophers working on the sciences of the mind, while Cluster 7 includes those specializing in the life sciences. Finally, on the eastern side of the map we find Cluster 4, consisting of philosophers of physics. This group appears relatively isolated from the rest of the network, suggesting that philosophers of physics form a tightly knit sub-community with limited connections to the broader field.

The map also illustrates the influence of professional associations (see the interactive version). Officials of the Philosophy of Science Association (PSA), the oldest and largest association, appear in all clusters, reflecting its broad reach across the field. By contrast, officials of the younger European Philosophy of Science Association (EPSA) are mainly concentrated in the general philosophy of science cluster, while officials of historically and practice-oriented associations (SPSP, HOPOS, & HPS) are, as expected, largely located in the integrated history and philosophy of science cluster.

Interestingly, our map shares many similarities with the fourth map of “hidden communities of interest” in philosophy of science produced by Malaterre and Lareau (2024), which is based on a comparable corpus of philosophy of science journals from a partly overlapping period (1996–2017), but relies on the topical profiles of authors—i.e., on similarities in their intellectual production—rather than on co-acknowledgment data.

First, both maps identify, alongside communities of philosophers of science, a group of analytic philosophers working on topics such as epistemology and philosophy of language (Cluster 1 in our map; communities a, b, and c in Malaterre and Lareau’s), highlighting the close connection between philosophy of science and analytic philosophy. Second, philosophers of science working on general philosophy of science using formal methods and probability theory (Cluster 3; communities d, e, and f) appear at the center of the map in both analyses. Third, philosophers of the life sciences (Cluster 7; community h) and philosophers of the sciences of the mind (Cluster 6; community i) are positioned close to each other in both maps. Fourth, philosophers of science integrating history of science (Cluster 2; community m) appear on both maps on the side opposite to analytic philosophy. Finally, philosophers of physics (Cluster 4; community k) emerge as a distinct and relatively isolated community in both networks.

These convergences suggest that, despite relying on different data sources and similarity measures, acknowledgment-based social networks and text-based intellectual similarity networks recover broadly comparable large-scale structures in contemporary philosophy

of science, thereby supporting the use of SOPHIS as a meaningful basis for exploring how social organization intersects with other dimensions of the field.

Future developments

SOPHIS can be further developed to address some of its current limitations. In this section, we outline possible improvements in three areas: field delineation, data collection, and data enrichment.

With respect to field delineation, the current version of SOPHIS does not cover books and collected volumes, even though these outlets continue to play an important role in the communication system of philosophy of science—despite the key place of journals that we noted above. Books and collected volumes, however, pose specific challenges, particularly with respect to accessing digitized full texts for collecting acknowledgments and indexing cited references. Moreover, SOPHIS could in the future be extended to cover sources in languages other than English. Although English is the dominant language of the field, important traditions in philosophy of science have emerged in other languages, such as French historical epistemology (Vagelli, 2024). Another natural direction for expansion is temporal, ideally extending SOPHIS back to the beginnings of philosophy of science as a professionalized discipline. These expansions would yield a more inclusive and representative picture of the field.

Regarding data collection, SOPHIS could be expanded to include information on editorial board memberships of journals and book series. Serving as an editor is not only a sign of prestige within the community but also entails significant influence in shaping the field by gatekeeping the publication process (Baccini & Barabesi, 2011). In this sense, editors are key figures who operate at the interface between the social and semiotic layers. Future versions of SOPHIS will incorporate this type of data, which will require the development of specific protocols for systematic collection. Furthermore, future versions may support linkage to open citation data retrieved from open bibliographic infrastructures such as OpenAlex.

Finally, SOPHIS could be further enriched with additional information on the semantic layer of philosophy of science. For example, topic modeling could be applied to the full text of articles to generate topics and assign them to papers, or actors could be associated with existing topical profiles such as those generated by Malaterre and Lareau. Alternatively, fuzzy clustering techniques could be employed, allowing overlaps between subfields. Articles could also be linked via their DOI to curated classification schemes such as those offered by PhilPapers, the Philosopher's Index, and the Internet Philosophy Ontology. Such interoperability would facilitate cross-infrastructure analyses that combine semantic classifications with social and institutional metadata.

Taken together, these extensions would further enhance SOPHIS as a comprehensive infrastructure for quantitative research on philosophy of science.

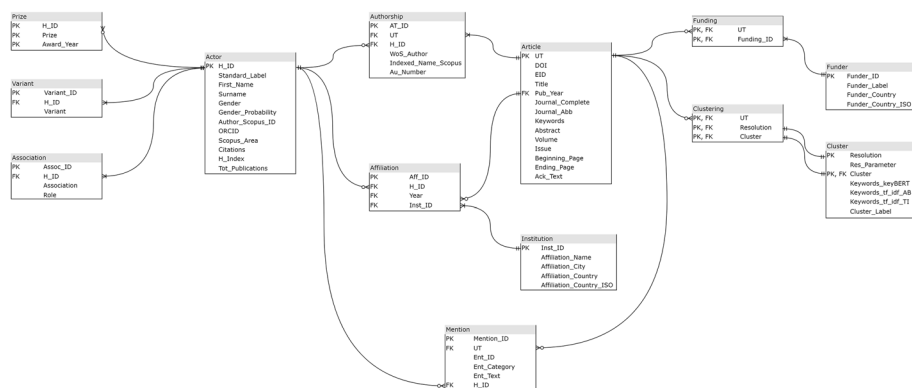


Fig. 4 Relational schema diagram of SOPHIS relational version

Conclusions

Quantitative methods are powerful tools for investigating the structure and dynamics of science and scholarship. To fully realize their potential, however, they require reliable, rich, and high-quality data. In the humanities, existing data sources still suffer from important limitations that constrain progress in quantitative research.

In this paper, we introduced SOPHIS, an open-access graph database designed for the quantitative study of contemporary philosophy of science. By integrating information on epistemic contents, their material carriers, and the individual and collective actors involved in their production, SOPHIS provides a comprehensive framework for mapping the structure, dynamics, and social foundations of the field.

We demonstrated SOPHIS’s potential through three illustrative proof-of-concept applications: identifying key actors at individual and institutional levels and examining the distribution of prestige, analyzing the factors influencing prestige accumulation, and mapping the fine-grained structure of communities within the field. Finally, we outlined several promising directions for further development of SOPHIS.

With SOPHIS, we hope to have contributed to building a more robust, open-access data infrastructure for the humanities—one that can support future generations of quantitative research on their social and intellectual dynamics.

Appendix

SOPHIS: relational database version

The relational version of SOPHIS comprises 13 tables. Its relational schema diagram is shown in Fig. 4.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11192-026-05591-4>.

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Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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